



Events in SQL

An **Event** in SQL (MySQL Event Scheduler) is a database object that executes a block of SQL code automatically based on a **time schedule**.

Unlike triggers (which are event-driven by INSERT/UPDATE/DELETE), **events are time-driven automation tools**.

They are commonly used for:

- Scheduled cleanup tasks
- Automated reporting
- Data archiving
- Periodic maintenance jobs
- ETL-style workflows

Key Difference: Event vs Trigger

Feature	Trigger	Event
Execution type	Data change (INSERT/UPDATE/DELETE)	Time schedule
Activation	Automatic on table action	Automatic on time interval
Scope	Row-level operations	Database-wide jobs
Use case	Audit, validation	Scheduling, maintenance

TechSphere Database Event Implementation

Below is a structured implementation using your **TechSphere Solutions Database System**.

1. Enable Event Scheduler

```
SET GLOBAL event_scheduler = ON;
```

What it does

- Activates MySQL event scheduler engine
- Required for all events to run



2. Event: Daily Salary Audit Summary

```
CREATE TABLE IF NOT EXISTS salary_summary_log (  
    log_id INT AUTO_INCREMENT PRIMARY KEY,  
    total_employees INT,  
    total_payroll DECIMAL(12,2),  
    average_salary DECIMAL(12,2),  
    log_date DATETIME  
);
```

EVENT: daily_salary_summary_event

PURPOSE: Generate daily payroll summary snapshot

Event Script

```
DELIMITER $$  
CREATE EVENT daily_salary_summary_event  
ON SCHEDULE EVERY 1 DAY  
STARTS CURRENT_TIMESTAMP  
DO  
BEGIN  
  
    INSERT INTO salary_summary_log (  
        total_employees,  
        total_payroll,  
        average_salary,  
        log_date  
    )  
    SELECT  
        COUNT(e.employee_id),  
        SUM(jr.base_salary),  
        AVG(jr.base_salary),  
        NOW()  
    FROM employees e  
    JOIN employee_positions ep ON e.employee_id = ep.employee_id  
    JOIN job_roles jr ON ep.role_id = jr.role_id;  
END $$  
DELIMITER ;
```



What it does

Creates a backup/archive table for deleted employee records.

Purpose

- Preserves deleted data before permanent removal.
- Supports recovery and historical reporting.

Every day:

- Calculates total employees
- Calculates total payroll cost
- Calculates average salary
- Stores snapshot in salary_summary_log

3. Event: Automatic Cleanup of Old Audit Logs

```
DELIMITER $$

CREATE EVENT cleanup_audit_logs
ON SCHEDULE EVERY 1 WEEK
DO
BEGIN

    DELETE FROM employee_audit_log
    WHERE action_timestamp < NOW() - INTERVAL 90 DAY;

END $$

DELIMITER ;
```

EVENT: cleanup_audit_logs

PURPOSE: Remove old logs older than 90 days

What it does

- Runs weekly
- Deletes old audit logs
- Keeps database optimized



4. Event: Archive Old Employees Automatically

```
DELIMITER $$

CREATE EVENT archive_old_employees
ON SCHEDULE EVERY 1 MONTH
DO
BEGIN

    INSERT INTO deleted_employees_archive
    (
        employee,
        first_name,
        last_name,
        gender,
        date_of_birth,
        hire_date,
        department_id,
        deleted_at
    )

    SELECT
        employee_id,
        first_name,
        last_name,
        gender,
        date_of_birth,
        hire_date,
        department_id,
        NOW()

    FROM employees
    WHERE hire_date < DATE_SUB(NOW(), INTERVAL 5 YEAR);

END $$

DELIMITER ;
```

EVENT: archive_old_employees

PURPOSE: Move old employees to archive table



What it does

- Runs monthly
- Moves employees older than 5 years into archive
- Helps historical data storage

5. How Events and Triggers Work Together

This is where database automation becomes powerful.

Example Workflow: Employee Lifecycle System

STEP 1: INSERT TRIGGER (Real-time action)

AFTER INSERT ON employees

What happens

- New employee added
- Trigger logs into employee_audit_log

STEP 2: UPDATE TRIGGER (Change tracking)

AFTER UPDATE ON employees

What happens

- Salary or department changes
- Trigger logs modifications

STEP 3: DELETE TRIGGER (Safety layer)

BEFORE DELETE ON employees



What happens

- Employee deleted
- Data copied to archive table first

STEP 4: EVENT (Scheduled automation)

ON SCHEDULE EVERY 1 DAY

What happens

- System generates payroll summaries daily
- Cleans old logs weekly
- Archives long-term employees monthly

Integrated System Flow

USER ACTIONS



INSERT / UPDATE / DELETE



TRIGGERS EXECUTE IMMEDIATELY



Audit logs + Archives updated

SYSTEM TIME EVENTS



Daily / Weekly / Monthly Jobs



Reports + Cleanup + Archiving



Business Use Case in TechSphere

HR & Payroll System

Component	Role
Triggers	Track employee changes in real time
Events	Generate reports & clean old data
Audit table	Compliance tracking
Archive table	Historical storage

Final Summary

Event Definition

An **Event** is a scheduled SQL task that runs automatically at defined time intervals.

In TechSphere System:

- **Triggers** handle real-time database changes
- **Events** handle scheduled automation tasks
- Together they create a:
 - Self-maintaining
 - Auditable
 - Automated HR & payroll system